Marco Piña

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Research Areas

Quantitative Macroeconomics, Financial Econometrics

Education

PhD Economics at **University of Virginia** 2023 - 2028 (expected)
MA Economics at **University of Virginia** 2023 - 2024
MS Economics at **University of Talca** 2018 - 2020

Work Experience

Research Assistant - EconTAI **Nov 2025 - present**

- Built an economics-knowledge benchmark comparing Claude, Gemini, and ChatGPT.
- Designed and implemented in Python an LLM research assistant using RAG and agentic workflows for literature reviews, mathematical derivations, and coding.
- Built end-to-end demonstrations for Prof. Korinek's course on AI agents.

Teaching Assistant - University of Virginia **Aug 2024 - Dec 2025**

- Money and Banking - Fall 2025
- Introduction to Macroeconomics - Spring 2025
- Introduction to Microeconomics - Fall 2024

Economist - Central Bank of Chile **Dec 2019 - Aug 2023**

At the Macrofinancial Modeling Department, I developed, calibrated, and estimated structural models with financial frictions, while also generating and managing databases for economic analysis. I collaborated with senior economists as both a research assistant and co-author on several projects.

Research Assistant - Project Fondecyt Regular N°1180672 **Mar 2019 - Dec 2019**

As a research assistant, I studied the role of option-implied volatilities in market risk modeling, designing and estimating a multivariate volatility model using financial options. The findings were documented in a working paper as part of the project.

Analyst (Intern) - Financial Market Commission **Jan 2019 - Mar 2019**

During my internship, I conducted literature reviews on stress-testing models for the Chilean banking system, worked with administrative data on market and credit risk, and reported results directly to the department.

Publications

Alfaro, Rodrigo and Marco Piña (2023). "Estimates of the US shadow-rate". In: *Latin American Journal of Central Banking* 4.1, p. 100080.

Jara, Alejandro and Marco Piña (2023). "Exchange rate volatility and the effectiveness of FX interventions: The case of Chile". In: *Latin American Journal of Central Banking* 4.2, p. 100086.

Herrera, Rodrigo and Marco Piña (2024). "Market risk modeling with option-implied covariances and score-driven dynamics". In: *The North American Journal of Economics and Finance* 72, p. 102136.

Working Papers

"Simple Implementable Financial Policy Rules" - with M. Calani and J. Moreno, [Draft]

Work in Progress

"The Sovereign-Bank Nexus, Credit Cycles and Macroprudential Policy" [Abstract]
"The Credibility Cost of Redemption: Local Currency Debt and the Inflation Risk Premium" [Abstract]

Awards

Snavely Distinguished Second Year Paper Prize (2025) - University of Virginia

Best Graduate of the Master's Program in Economics (2020) - University of Talca

Research and Computational Skills

Generative AI	LLM evaluation, retrieval-augmented generation, and agentic research workflows
Developer Tools	Git/Github, VS Code, Jupyter, SQL
Programming	Julia, Python, R
Deep Learning	Flux, PyTorch
Softwares	Stata, Matlab/Dynare
Datastream	Bloomberg, Eikon-Thomson Reuters
Languages	Spanish (Native), English (Proficient)

Courses and Training

"Machine Learning in Finance" - CEMFI	Aug 2022
"Introduction to Mathematical Thinking" - Stanford Online	Jul 2021
"Continuous Time in Macroeconomics with Machine Learning" - University of Oxford	Jun 2021
"Numerical Methods for Mathematics with Octave" - UPValenciaX	Feb 2021
"Occasionally Binding Constraints in Macroeconomics" - Carleton-Bank of Canada	Jan 2021
"Monetary Policy Analysis and Forecasting" - IMFx	Jan 2021
"Tools for Macroeconomists" - University of Oxford	Aug 2020